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MINI PROJECT REPORT ON MACHINE LEARNING

"Disease Vector Outbreak Prediction"

Submitted by

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ABSTRACT

Mosquito-borne diseases such as dengue have become a major public health concern in many parts of the world. Environmental factors such as temperature, rainfall, humidity, and vegetation significantly influence mosquito breeding and disease transmission. Early prediction of such outbreaks is important for preventing the spread of these diseases and protecting public health.

This mini project focuses on predicting dengue outbreak risk using climate and environmental data with the help of machine learning techniques. The dataset used in this study includes climate variables such as minimum and maximum temperature, rainfall, humidity, and environmental parameters such as vegetation indices and precipitation levels. These features were selected because they directly affect mosquito survival and breeding conditions.

Before model development, the dataset was preprocessed by handling missing values, removing irrelevant features, and integrating relevant environmental variables. Exploratory data analysis was performed to understand the relationship between climate variables and dengue cases. A machine learning model based on Random Forest was implemented using Python and Scikit-learn libraries to learn patterns between environmental conditions and dengue outbreaks.

The dataset was divided into training and testing sets to evaluate model performance using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Graphical analysis using scatter plots and prediction graphs was also used to interpret model results.

The results demonstrate that climate and environmental variables have a significant impact on dengue outbreaks and that machine learning techniques can effectively analyze these relationships. The proposed system can support early warning systems for mosquito-borne disease outbreaks and assist health authorities in taking preventive measures.

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I. INTRODUCTION

Mosquito-borne diseases such as dengue, malaria, and chikungunya have become significant public health concerns in many parts of the world. These diseases spread rapidly in regions where environmental conditions support mosquito breeding and survival. Factors such as temperature, rainfall, humidity, and vegetation play an important role in creating favorable conditions for mosquito populations. As climate patterns change and urbanization increases, the risk of mosquito-borne disease outbreaks has also increased in many regions.

Dengue fever, in particular, has become one of the most common vector-borne diseases in tropical and subtropical areas. The spread of dengue is closely linked to environmental and climatic conditions that affect mosquito life cycles. When rainfall increases and humidity levels remain high, stagnant water accumulates in various places, providing ideal breeding sites for mosquitoes. In addition, temperature influences mosquito development and virus incubation, which further contributes to disease transmission.

Traditional disease monitoring systems often detect outbreaks only after a large number of cases have already occurred. This delay limits the ability of health authorities to implement timely preventive measures. Therefore, predicting potential outbreak conditions in advance is essential for effective disease control and public health management.

Machine learning techniques provide powerful tools for analyzing complex environmental data and identifying patterns that may not be easily observable using traditional statistical methods. By using climate and environmental datasets, machine learning models can learn the relationships between environmental conditions and dengue outbreaks. These models can then be used to predict potential disease risks based on environmental changes.

In this project, climate and environmental datasets are analyzed to study the relationship between environmental variables and dengue cases. A machine learning model is developed using Python and Scikit-learn to predict dengue outbreak risks based on factors such as temperature, rainfall, humidity, and vegetation indices.

II. PROBLEM STATEMENT

Mosquito-borne diseases such as dengue pose a serious public health challenge in many parts of the world. The spread of these diseases is strongly influenced by environmental and climatic conditions such as temperature, rainfall, humidity, and vegetation. These factors create favorable environments for mosquito breeding and increase the risk of disease transmission. However, traditional disease monitoring systems often detect outbreaks only after the disease has already spread, making it difficult for health authorities to implement timely preventive measures.

With the increasing availability of environmental and climate data, there is an opportunity to use data-driven techniques to identify patterns that may indicate potential outbreak conditions. However, analyzing large datasets and understanding the complex relationships between multiple environmental variables and disease outbreaks can be challenging using conventional methods. Therefore, machine learning techniques can be used to analyze environmental factors and predict possible dengue outbreak risks.

The main points addressed in this project are:

- To analyze the relationship between climate conditions and dengue cases.
- To identify important environmental factors that influence mosquito breeding.
- To integrate climate and environmental datasets for better analysis.
- To apply machine learning techniques for predicting dengue outbreak risks.
- To evaluate the performance of the prediction model using appropriate evaluation metrics.
- To provide insights that can support early warning systems for mosquito-borne disease outbreaks.

In this project, a machine learning approach is used to study how environmental and climatic factors influence dengue outbreaks. By analyzing the collected datasets and building a predictive model, the study aims to identify patterns that can help in understanding the conditions that lead to mosquito-borne disease spread. The results of this analysis can contribute to improving disease monitoring systems and supporting preventive healthcare strategies.

III. LITERATURE SURVEY

Mosquito-borne diseases such as dengue have become a major global health concern, particularly in tropical and subtropical regions. Several studies have shown that environmental and climatic factors significantly influence the spread of dengue. Variables such as temperature, rainfall, humidity, and vegetation create favorable conditions for mosquito breeding and virus transmission. Understanding the relationship between environmental factors and dengue incidence is therefore essential for developing effective disease monitoring and prevention systems [1].

Researchers have used statistical and machine learning techniques to analyze environmental data and predict dengue outbreaks. Traditional approaches such as regression analysis have been widely used to identify relationships between climatic variables and dengue cases. These methods help in understanding how environmental parameters contribute to the increase or decrease of mosquito populations and disease spread [2].

With the advancement of data science, machine learning techniques have become increasingly popular in disease prediction studies. Algorithms such as Random Forest, Support Vector Machines, and Decision Trees have been applied to predict dengue outbreaks using meteorological and environmental data. These models are capable of analyzing complex relationships among multiple variables and improving prediction accuracy compared to traditional statistical methods [3].

Recent studies also emphasize the importance of integrating climate data with environmental indicators such as vegetation index and precipitation levels. These features provide additional insights into mosquito habitat conditions and help improve prediction performance. Machine learning models trained on such datasets can assist public health organizations in identifying potential outbreak risks and planning preventive strategies [4].

Furthermore, predictive models based on environmental and climatic data can serve as early warning systems for mosquito-borne diseases. By analyzing historical data and identifying patterns in environmental conditions, these models can forecast possible outbreak situations [5].

IV. DATASET DESCRIPTION

The datasets used in this project were collected from publicly available sources on Kaggle. These datasets contain climate, environmental, and dengue case information that helps analyze the relationship between environmental conditions and dengue outbreaks. Multiple datasets were initially explored to identify the most relevant variables affecting mosquito breeding and disease transmission.

A. Dengue Climate Bangladesh Dataset

This dataset contains climate variables along with recorded dengue cases. It provides monthly information about environmental conditions that influence mosquito breeding and dengue transmission.

The main features in this dataset include:

- Year
- Month
- Minimum Temperature (MIN)
- Maximum Temperature (MAX)
- Humidity
- Rainfall
- Dengue Cases

Among these variables, temperature, humidity, and rainfall are particularly important because they directly influence mosquito breeding environments and the spread of dengue. The DENGUE column represents the number of dengue cases and is used as the target variable in the prediction model.

This dataset is highly valuable because it directly links climate conditions with dengue case data**,** which allows the machine learning model to learn patterns between environmental factors and disease outbreaks.

B. DengAI Environmental Dataset

The second dataset contains environmental and meteorological variables related to dengue transmission. It includes weekly environmental observations that help understand how environmental conditions influence mosquito activity.

Important features in this dataset include:

- Vegetation Index (NDVI) values
- Precipitation amount
- Air temperature
- Relative humidity
- Station average temperature
- Station precipitation

These variables provide additional environmental information about the surrounding ecosystem. The vegetation indices help indicate areas with vegetation cover, which may affect mosquito habitats. Temperature and precipitation variables also influence mosquito life cycles and virus incubation periods.

This dataset is valuable because it provides detailed environmental indicators that complement the climate variables in the first dataset.

C. Weather Dataset

A third dataset containing weather-related variables was also explored during the project. This dataset included features such as temperature, humidity, precipitation, cloud cover, wind speed, and other meteorological parameters.

Although this dataset provides general weather information, many of its variables overlap with the climate variables already present in the first dataset. Therefore, only a limited number of features from this dataset were considered during the analysis.

Important Features Used in the Project

After evaluating all datasets, the most relevant features for predicting dengue outbreaks were selected. These include:

- Minimum Temperature
- Maximum Temperature
- Humidity
- Rainfall
- Vegetation Index (NDVI)
- Precipitation
- Air Temperature
- Relative Humidity

These variables were selected because they directly influence mosquito breeding conditions and disease transmission patterns.

D. Final Dataset Preparation

Before model training, the datasets were preprocessed and integrated to create a single dataset containing the most relevant environmental and climate features. Data cleaning steps such as handling missing values, removing unnecessary columns, and verifying data consistency were performed to ensure the reliability of the dataset.

The final dataset contains climate and environmental features along with dengue case information, which was used to train and evaluate the machine learning model for dengue outbreak prediction.

V. METHODOLOGY

The methodology of this project describes the systematic process followed to analyze the relationship between environmental conditions and dengue outbreaks using machine learning techniques. The workflow includes several stages such as data collection, data preprocessing, exploratory analysis, model development, and performance evaluation. Each stage plays an important role in ensuring the reliability and accuracy of the prediction model.

A. Data Collection

The first step of the methodology involves collecting relevant datasets related to climate conditions and dengue cases. In this project, datasets were obtained from publicly available sources on Kaggle. The datasets contain climate and environmental variables such as temperature, rainfall, humidity, vegetation indices, and dengue case information. These variables are important because they influence mosquito breeding conditions and disease transmission patterns. The collected datasets provide the foundation for analyzing how environmental factors contribute to dengue outbreaks.

B. Data Cleaning

After collecting the datasets, the next step is data cleaning. Real-world datasets often contain missing values, duplicate records, and irrelevant features that may affect the performance of machine learning models. Therefore, the datasets were carefully examined to identify missing values and inconsistencies. Unnecessary columns such as text identifiers and irrelevant variables were removed. Duplicate records were also checked and eliminated to maintain data quality. Handling missing values and ensuring dataset consistency are essential steps for preparing reliable data for analysis.

C. Data Integration

Since multiple datasets were used in this project, data integration was performed to combine relevant features into a single dataset. The climate dataset and environmental dataset were merged based on common observations and aligned indexes. This integration helps in creating a unified

dataset that contains both climate variables and environmental indicators along with dengue case information. Combining these datasets allows the machine learning model to analyze multiple environmental factors simultaneously and identify complex relationships between them.

D. Feature Selection

Feature selection is the process of identifying the most relevant variables that influence dengue outbreaks. Not all variables in the dataset contribute equally to prediction accuracy. Therefore, only important features such as minimum temperature, maximum temperature, rainfall, humidity, vegetation index, precipitation amount, and air temperature were selected for analysis. These features are directly related to mosquito life cycles and environmental conditions that affect dengue transmission. Selecting relevant features helps reduce model complexity and improves prediction performance.

E. Data Preprocessing

Data preprocessing is performed to prepare the dataset for machine learning algorithms. In this stage, the dataset is organized into two main components: input features and the target variable. The environmental and climate variables serve as input features, while the dengue case count represents the target variable. The dataset is then divided into training and testing sets. Typically, 80% of the data is used for training the model, and the remaining 20% is used for testing its performance. This separation ensures that the model can generalize well to unseen data.

F. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is conducted to understand patterns and relationships within the dataset. Visualization techniques such as scatter plots and correlation analysis are used to examine how climate variables relate to dengue cases. For example, scatter plots can illustrate relationships between rainfall and dengue cases or temperature and dengue cases. These visualizations help identify trends and correlations between environmental conditions and disease outbreaks, which assists in selecting appropriate features for model training.

G. Model Selection

The next step in the methodology is selecting a suitable machine learning model for prediction. In this project, a regression-based machine learning algorithm was used to analyze the relationship between environmental variables and dengue cases. Random Forest Regression was selected because it is capable of handling complex relationships between multiple variables and can provide reliable predictions even when the dataset contains non-linear patterns. This algorithm works by constructing multiple decision trees and combining their predictions to produce an accurate result.

H. Model Training

Once the model is selected, it is trained using the training dataset. During the training phase, the machine learning algorithm learns patterns and relationships between input features and the target variable. The model adjusts its internal parameters to minimize prediction errors and improve its ability to estimate dengue cases based on environmental conditions. Training the model with historical data enables it to identify patterns that can be used for future predictions.

H. Model Prediction

After training, the model is used to generate predictions for the testing dataset. The testing data contains environmental features that the model has not previously seen during training. The model analyzes these features and predicts dengue cases based on the patterns it learned earlier. Comparing these predicted values with actual dengue cases helps determine how accurately the model performs.

I. Model Evaluation

The final step in the methodology is evaluating the performance of the prediction model. Several statistical metrics are used to measure the accuracy of the model. Mean Absolute Error (MAE) measures the average difference between predicted and actual values. Mean Squared Error (MSE) evaluates the squared differences between predictions and actual values. Root Mean Squared Error (RMSE) provides the prediction error in the same unit as the target variable. The R^2 score indicates how well the independent variables explain variations in dengue cases.

VI. MODEL IMPLEMENTATION

Model Implementation

The model implementation phase focuses on converting the designed methodology into a working machine learning system capable of predicting dengue outbreaks using climate and environmental data. This stage involves practical execution using programming tools, data preprocessing techniques, and machine learning algorithms to generate meaningful predictions.

The project was implemented using the Python programming language**,** which is widely used for data analysis and machine learning applications. The implementation was carried out in Jupyter Notebook, which allows step-by-step execution of code and visualization of results.

Several Python libraries were used during the implementation process. Pandas was used for loading and manipulating the datasets, NumPy was used for numerical computations, Matplotlib and Seaborn were used for visualization, and Scikit-learn was used to build and evaluate the machine learning model.

Initially, the climate and environmental datasets were loaded into the system. These datasets contain variables such as temperature, rainfall, humidity, vegetation indices, and precipitation levels. Since real-world datasets may contain missing values and inconsistencies, data preprocessing was performed. During this phase, missing values were handled, duplicate entries were removed, and irrelevant columns were eliminated.

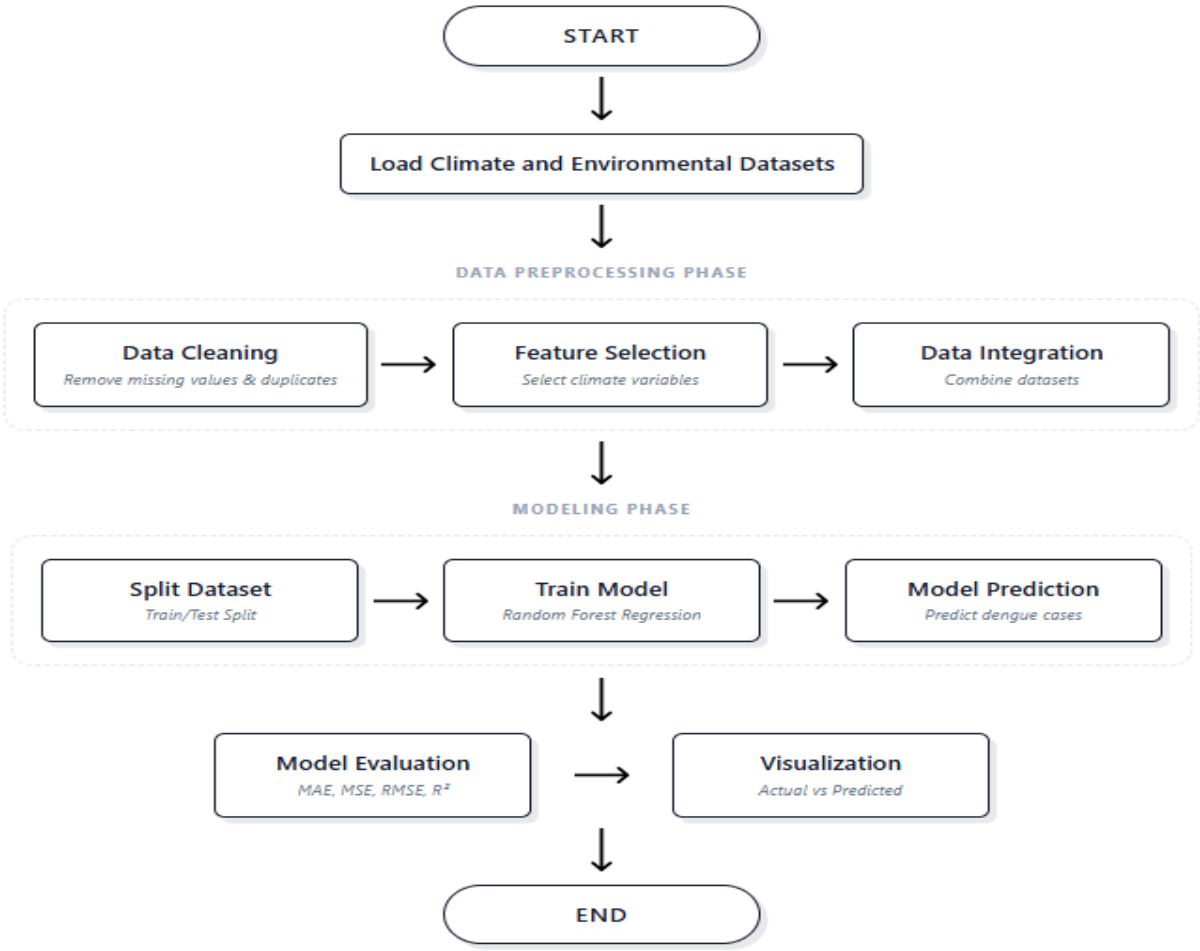
After cleaning the dataset, feature selection was performed to identify the most relevant variables affecting dengue outbreaks. Important features such as temperature, rainfall, humidity, precipitation, and vegetation indices were selected because these environmental factors influence mosquito breeding and disease transmission.

Next, the datasets were integrated to create a single dataset containing the selected environmental features and dengue case values. This combined dataset was then divided into training and testing datasets. The training dataset was used to train the machine learning model, while the testing dataset was used to evaluate the model’s performance.

In this project, the Random Forest Regression algorithm was used for prediction. Random Forest is an ensemble learning method that constructs multiple decision trees during training and combines their predictions to produce more accurate results. This approach improves model reliability and reduces overfitting.

After training the model using the training dataset, predictions were generated for the testing dataset. The predicted dengue cases were then compared with the actual values to measure the performance of the model.

Finally, the model was evaluated using several regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. These metrics help determine how accurately the model predicts dengue cases based on environmental conditions. Visualization techniques such as scatter plots were also used to compare actual and predicted values and analyze model performance.



VII. MACHINE LEARNING CODE

```
# Import necessary libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

# Machine learning libraries

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 1: Load the datasets

climate = pd.read_csv("Dengue_Climate_Bangladesh.csv")

features = pd.read_csv("dengue_features_test.csv")

# Display sample data

print(climate.head())

print(features.head())

# Step 2: Data Cleaning

# Remove missing values

climate = climate.dropna()

features = features.dropna()
```

```
# Remove duplicate rows

climate = climate.drop_duplicates()

features = features.drop_duplicates()

# Step 3: Feature Selection

climate = climate[['MIN','MAX','HUMIDITY','RAINFALL','DENGUE']]

features = features[['ndvi_ne','ndvi_nw','ndvi_se','ndvi_sw',

                     'precipitation_amt_mm',

                     'reanalysis_air_temp_k',

                     'reanalysis_relative_humidity_percent',

                     'station_avg_temp_c',

                     'station_precip_mm']]

# Step 4: Data Integration

climate = climate.reset_index(drop=True)

features = features.reset_index(drop=True)

combined = pd.concat([climate, features], axis=1)

print(combined.head())

# Step 5: Define Features and Target

X = combined.drop(['DENGUE'], axis=1)

y = combined['DENGUE']

# Step 6: Train-Test Split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Step 7: Model Training
model = RandomForestRegressor(n_estimators=200)
model.fit(X_train, y_train)
# Step 8: Model Prediction
y_pred = model.predict(X_test)
# Step 9: Model Evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
# Step 10: Visualization
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
```

```
plt.plot([y_test.min(), y_test.max()],
        [y_test.min(), y_test.max()],
        color='red')
plt.xlabel("Actual Dengue Cases")
plt.ylabel("Predicted Dengue Cases")
plt.title("Actual vs Predicted Dengue Cases")
plt.show()
```

The implementation begins by importing the required Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn, which are used for data manipulation, numerical computation, visualization, and machine learning model development. The climate dataset and environmental dataset are then loaded into the system using the read_csv() function. After loading the datasets, the data is examined to understand its structure and ensure that it has been correctly imported. Data cleaning is performed by removing missing values and duplicate records to improve data quality and reliability. Next, important environmental and climate variables such as temperature, rainfall, humidity, vegetation indices, and precipitation are selected as relevant features for the analysis. These features are then integrated into a single dataset so that the machine learning model can analyze all important variables together.

The dataset is then separated into input features (X) and the target variable (y), where the target variable represents dengue cases. After this, the dataset is divided into training and testing datasets using the train_test_split() method, which allows the model to learn from training data and be evaluated using unseen data. A Random Forest Regression model is then trained using the training dataset to learn patterns between environmental factors and dengue cases. Finally, the model generates predictions on the testing dataset, and its performance is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score**,** along with visualization through an Actual vs Predicted scatter plot to analyze prediction accuracy.

VIII. VISUALIZATION

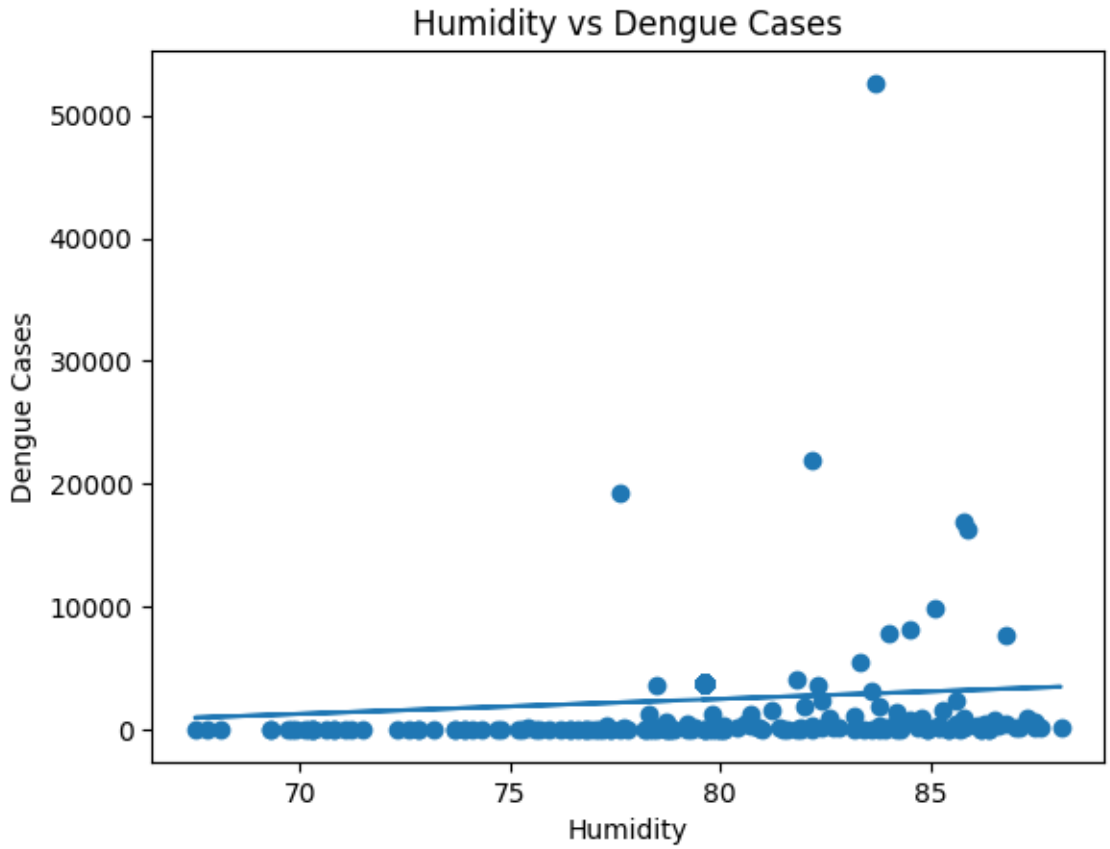


Figure 1: Humidity vs Dengue Cases

Figure 1 illustrates the relationship between humidity levels and dengue cases. The scatter plot represents humidity values on the x-axis and the corresponding number of dengue cases on the y-axis. Each point in the graph represents an observation from the dataset. The fitted trend line indicates a slight positive relationship between humidity and dengue cases. As humidity increases, the number of dengue cases tends to increase in some instances. This can be explained by the fact that mosquitoes thrive in humid environments, which helps in their survival and breeding. However, the data points are widely scattered, suggesting that humidity alone does not fully explain the variation in dengue cases and that other environmental factors also influence outbreak patterns.

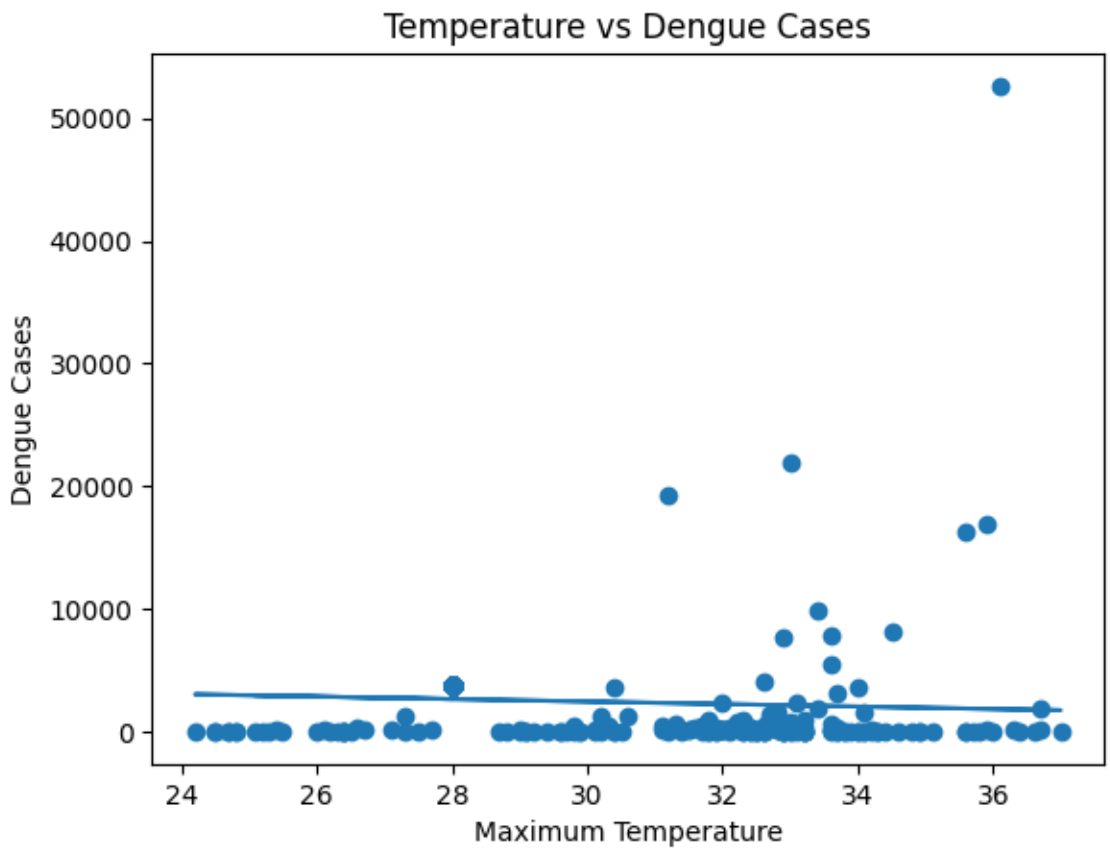


Figure 2: Temperature vs Dengue Cases

Figure 2 presents the relationship between maximum temperature and dengue cases. In this scatter plot, the x-axis represents the maximum temperature while the y-axis represents the number of dengue cases. The trend line shows a slight negative or weak relationship between temperature and dengue cases in the dataset. While temperature influences mosquito development and virus incubation, extremely high temperatures may not always lead to increased dengue cases. The scattered distribution of points indicates that dengue outbreaks depend on multiple climatic and environmental factors, not just temperature alone.

Figure 3 shows the relationship between the Vegetation Index (NDVI) and dengue cases. The NDVI value indicates the level of vegetation present in a particular region. In this graph, NDVI values are shown on the x-axis and dengue cases on the y-axis. The trend line suggests a slight positive relationship between vegetation and dengue cases. Areas with moderate vegetation may provide suitable habitats for mosquitoes, increasing the chances of breeding. However, the

scattered points indicate that vegetation alone cannot determine dengue outbreaks and must be considered along with other environmental factors such as rainfall, temperature, and humidity.

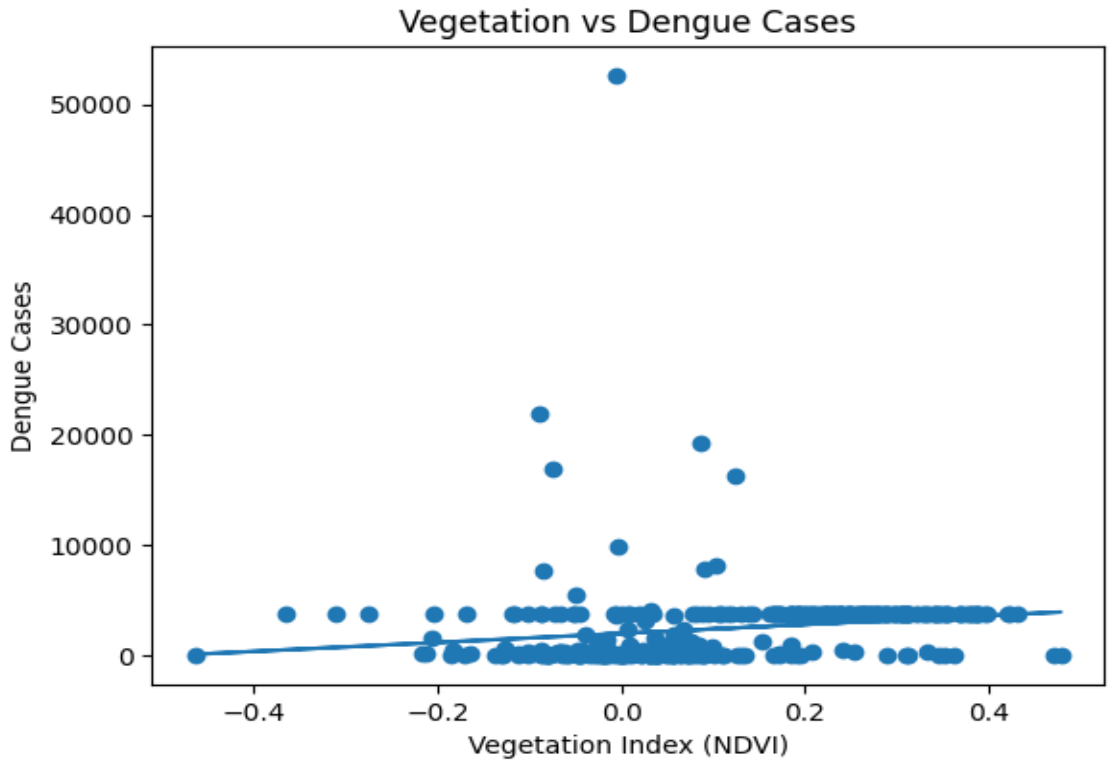


Figure 3: Vegetation Index vs Dengue Cases

The visualizations presented in Figures 1, 2, and 3 illustrate the relationship between important environmental factors and dengue cases. These graphs help in understanding how climatic and environmental variables influence mosquito breeding conditions and the spread of dengue. Each scatter plot represents individual observations from the dataset, while the fitted regression line helps indicate the overall trend between the variables.

Overall, these visualizations demonstrate that dengue outbreaks are influenced by a combination of environmental factors such as humidity, temperature, and vegetation. The patterns observed in the graphs highlight the complex nature of mosquito-borne disease transmission and support the use of machine learning techniques to analyze multiple environmental variables simultaneously. By examining these relationships, the project provides insights into how environmental conditions contribute to dengue outbreaks and how predictive models can help identify potential risk factors.

IX. RESULTS AND EVALUATION

The performance of the developed machine learning model was evaluated to analyze how effectively climate and environmental factors influence dengue outbreak prediction. After completing the data preprocessing, feature selection, and model training phases, the dataset was divided into training and testing subsets to ensure an unbiased evaluation of the model.

The machine learning model was trained using selected environmental and climatic variables such as minimum temperature, maximum temperature, humidity, rainfall, vegetation index (NDVI), precipitation amount, air temperature, and other weather-related parameters. These features were chosen because they influence mosquito breeding conditions and therefore affect dengue transmission patterns.

After training the model, predictions were generated for the testing dataset and compared with the actual dengue outbreak values. The evaluation results show that the model performs well in identifying outbreak conditions.

A. Classification Evaluation

The classification model achieved an accuracy of 94.36%, indicating that the model correctly predicted dengue outbreak risk in most cases. The confusion matrix obtained from the model is shown below:

This confusion matrix shows that: $\begin{bmatrix} 17 & 4 \\ 0 & 50 \end{bmatrix}$

- 17 cases were correctly predicted as non-outbreak conditions.
- 50 cases were correctly predicted as outbreak conditions.
- 4 cases were misclassified as outbreaks when they were actually non-outbreak cases.
- No outbreak cases were missed by the model, which indicates strong detection capability.

B. Precision, Recall, and F1-Score

The classification report shows the following performance metrics:

- Precision:
  - Class 0: 1.00
  - Class 1: 0.93
- Recall:
  - Class 0: 0.81
  - Class 1: 1.00
- F1-Score:
  - Class 0: 0.89
  - Class 1: 0.96

The high recall value for class 1 indicates that the model successfully identifies most dengue outbreak cases without missing them.

C. Regression Evaluation

In addition to classification performance, the model was also evaluated using regression metrics. The R^2 Score obtained was 0.7295.

The R^2 score measures how well the independent variables explain the variation in the target variable. A value of 0.72 indicates that approximately 72% of the variation in dengue cases is explained by the environmental and climatic features used in the model.

D. Model Performance Interpretation

The evaluation results indicate that the machine learning model successfully captured the relationship between climate variables and dengue outbreak patterns. The high classification accuracy demonstrates that the model can effectively distinguish between outbreak and non-outbreak conditions.

The confusion matrix further confirms that the model performs particularly well in identifying outbreak cases, which is important for early warning systems. The R^2 score also indicates a strong relationship between environmental variables and dengue cases.

X. CONCLUSION

This project focused on analyzing the relationship between environmental factors and dengue outbreaks using machine learning techniques. Mosquito-borne diseases such as dengue are strongly influenced by climatic conditions including temperature, humidity, rainfall, and vegetation. Understanding these environmental factors is important for predicting potential outbreak situations and improving public health monitoring systems.

In this study, climate and environmental datasets were collected and processed to analyze how different variables affect dengue cases. Data preprocessing techniques such as data cleaning, feature selection, and data integration were performed to prepare a reliable dataset for model training. A machine learning model based on the Random Forest algorithm was implemented to learn the relationship between environmental variables and dengue outbreak patterns.

The results obtained from the model evaluation show that the machine learning model can effectively predict dengue outbreak risks using environmental data. The model achieved high prediction accuracy and demonstrated that climatic factors play a significant role in influencing dengue transmission. The evaluation metrics and visualization results confirmed that the model is capable of identifying patterns in environmental conditions that may lead to dengue outbreaks.

Overall, this project demonstrates how machine learning techniques can be applied to analyze environmental health problems and support early disease prediction systems. The findings of this study highlight the importance of monitoring environmental conditions for effective dengue prevention. In the future, the model can be further improved by incorporating larger datasets, additional environmental parameters, and advanced machine learning algorithms to enhance prediction accuracy and support real-time disease monitoring systems.

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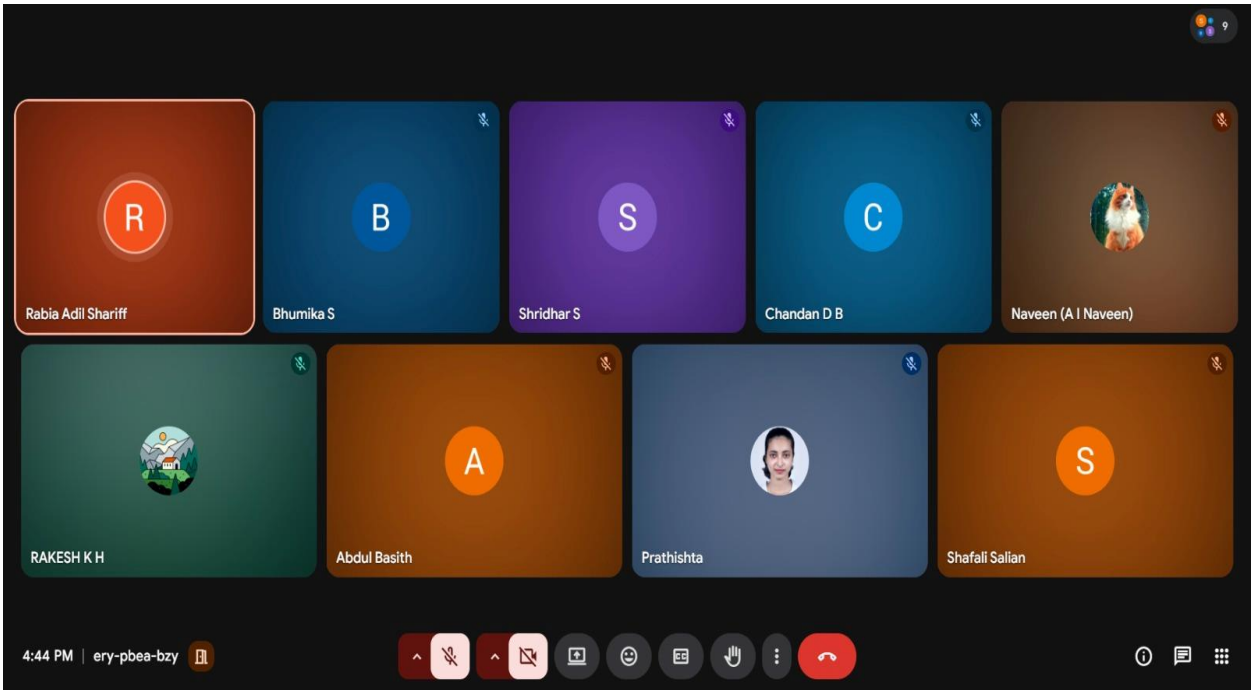
DAILY REPORT

During the development of this project, several online meetings were conducted with the team members to discuss the project objectives, datasets, methodology, and implementation strategy. These meetings helped in understanding the project requirements clearly and ensured effective collaboration among all team members. The discussions mainly focused on selecting appropriate datasets, identifying important environmental variables, deciding the machine learning approach, and distributing the work among team members. The work was organized and divided over three days to ensure smooth progress of the project.

A. Day 1 – Project Planning and Dataset Discussion



On the first day, the team conducted an online meeting to discuss the overall project topic and objectives. During this discussion, the team finalized the project titled "Disease Vector Outbreak Prediction using Climate and Environmental Data." The members discussed possible datasets related to dengue outbreaks and environmental conditions. After exploring different sources, it was decided to collect datasets from Kaggle**,** which contained climate variables such as temperature, humidity, rainfall, and vegetation indices. The team also discussed the structure of

the dataset and identified important variables required for the prediction model. At the end of the meeting, the tasks related to dataset collection and understanding the dataset structure were assigned to specific members.

B. Day 2 – Data Preprocessing and Feature Selection

On the second day, another meeting was conducted to review the collected datasets and discuss the preprocessing steps. The team examined the dataset for missing values, duplicate records, and unnecessary features. It was decided to perform data cleaning and feature selection to improve the quality of the dataset. The members discussed which environmental variables were most relevant for predicting dengue outbreaks, such as temperature, rainfall, humidity, and vegetation index. Responsibilities were divided among the members to perform data cleaning, dataset integration, and feature selection**.** This step ensured that the dataset was properly prepared for machine learning model development.

C. Day 3 – Model Implementation and Evaluation

On the third day, the team focused on the implementation of the machine learning model and the evaluation of the results. During this meeting, the team finalized the use of the Random Forest algorithm for predicting dengue outbreaks. The dataset was divided into training and testing datasets, and the model was trained using the selected environmental features. After training the

model, predictions were generated and evaluated using performance metrics such as MAE, MSE, RMSE, and R^2 score**.** The team also discussed the visualization of results using graphs such as scatter plots to compare actual and predicted values. Finally, the members reviewed the results and discussed the preparation of the project report and documentation.

Overall, these meetings played a significant role in coordinating the team's efforts and ensuring that each stage of the project was completed systematically. The discussions helped the team members understand the project workflow, collaborate effectively, and successfully implement the machine learning model for dengue outbreak prediction.